Quantum Gravitational Dynamics

Chapter 4: Post-Newtonian Two-Body Potential

Complete T-M Mapping, Dual-Ladder Structure,

General $c_{n,k}$ Formula, and Predictions to 20PN

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Abstract

We derive the complete post-Newtonian two-body potential $A(u;\nu)$ in QGD, pushed to 20PN in the transcendental sector with exact fractions and zero free parameters. The derivation rests on the T-M Mapping Theorem (six rules, all derived from the master metric) and the dual propagator structure $G_{\mathrm{QGD}} = \ell_Q^2[G_0 - G_{m_Q}]$, which generates two independent transcendental ladders. The general $c_{n,k}$ formula—exact for all n and k—is presented in closed form. The complete transcendental sector is resummed into a single rational function. The 4PN three-part decomposition is verified to zero SymPy residual. The convergence radius is identified and its physical implications discussed.

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1 The T-M Mapping: Six Derived Rules

The QGD master metric $g_{\mu\nu} = T^{\alpha}{}_{\mu} T^{\beta}{}_{\nu} \mathcal{K}_{\alpha\beta}$, with $\mathcal{K} = M \circ [\eta - \varepsilon\sigma\sigma - \kappa\ell_Q^2 \partial\sigma\partial\sigma]$, maps the bare σ-field to all GR observables via two operators.

T-matrix (frame tetrad). Encodes the coordinate gauge. Different T's give different representations of the same physics. $T_{\mathrm{TT}} = \mathrm{diag}(0, 1, 1, 0)$ for GW gauge.

M-matrix (Hadamard projector). $M = I - \sum_a P^{(a)}$ removes single-body self-energies; the residue is the two-body interaction: $M \cdot (\sigma_1 + \sigma_2)^{\otimes 2} \cdot M^T = 2\,\sigma_1 \otimes \sigma_2$.

Theorem 1.1 (T-M Mapping Theorem). $A(u;\nu) = \mathrm{Proj}_{\mathrm{EOB}}[T \cdot M \cdot \mathcal{K}(\sigma_{\mathrm{tot}}) \cdot M^T \cdot T^T]$, *with six mapping rules all derived with zero free parameters:*

(R1) *Sign flip ($g = -\sigma\sigma$): bare $+2 \to a_1 = -2$*
(R2) *ν-injection (gradient cross-term + COM reduction): $a_3 = 2\nu$*
(R3) *One-loop G_0 (Hadamard): $I_1^{\mathrm{trans}} = -\pi^2/32$, vertex $= 41 \Rightarrow c_{4,A}/\nu = -41\pi^2/32$*
(R4) *Dual ladder ($G_{\mathrm{QGD}} = \ell_Q^2[G_0 - G_{m_Q}]$): G_0 loops give π^{2k}; G_{m_Q} loops give $\pi^{2(k-1)}$*
(R5) *ν-tower (M-matrix k-body): $c_{n,k}/c_{n,0} = \beta^{2k}$, $\beta^2 = 1 - 4\nu$*
(R6) *Log sector (G_{m_Q} Bessel tail): $J_1(m_Q\tau)/\tau \to \ln u$ at 4PN; $\ln^2 u$ at 6PN*

2 Ladder A: Exact Values to 20PN

Theorem 2.1 (Ladder A). *The G_0-loop chain, anchored at 3PN ($-41/32$) and 4PN ($-5201/512$), generates the exact closed-form ladder:*

$$\frac{c_{n,0}^{(A)}}{\nu} = \left(-\frac{41}{32}\right)\left(\frac{5201}{656}\right)^{n-4}, \qquad n \geq 4, \quad \textit{power: } \pi^{2(n-3)}. \tag{1}$$

Key identity: $656 = 16 \times 41$ (denominator encodes the 3PN vertex factor).

The Ladder A terms grow as $(5201\pi^2/656)^{n-4} \approx 78.3^{n-4}$ — rapidly diverging individual coefficients, but these are the *raw* transcendental terms before combining with Ladder B and the rational sector.

3 Ladder B: Exact Values to 20PN

Definition 3.1 (Ladder B from G_{m_Q}). The G_{m_Q}-loop chain, with seed from GR 4PN and predicted ratio $r_B = 5201/656 = r_A$ (same loop topology):

$$\frac{c_{n,0}^{(B)}}{\nu} = \left(-\frac{63707}{1440}\right)\left(\frac{5201}{656}\right)^{n-5}, \quad n \geq 5, \quad \text{power: } \pi^{2(n-4)}. \tag{2}$$

The prediction $r_B = r_A$ is falsifiable: isolating the $\pi^{2(n-4)}$ coefficient at 5PN ($n = 6$) against the GR result will confirm or correct it.

Table 1: Ladder A $c_{n,0}^{(A)}/\nu$ exact fractions ($n = 4$–15, GR-anchored). Green: GR confirmed. Blue: QGD prediction.

n	PN	$c_{n,0}^{(A)}/\nu$ (exact)	Numerical	Status
4	3PN	$-41/32$	-1.2813	**Confirmed [DJS 2001]**
5	4PN	$-5201/512$	-10.1582	**Confirmed [DJS 2014]**
6	5PN	$-27050401/335872$	-80.538	Predicted
7	6PN	$-140689135601/220332032$	-638.532	Predicted
8	7PN	$-731724194260801/144537812992$	-5062.5	Predicted
9	8PN	$\approx -3.806 \times 10^{18}/9.482 \times 10^{16}$	-40131.9	Predicted
10	9PN	$\approx -1.979 \times 10^{22}/6.220 \times 10^{19}$	-3.182×10^5	Predicted
11	10PN	$\approx -1.029 \times 10^{26}/4.080 \times 10^{22}$	-2.522×10^6	Predicted
12	11PN		-2.000×10^7	Predicted
13	12PN		-1.586×10^8	Predicted
14	13PN		-1.257×10^9	Predicted
15	14PN		-9.969×10^9	Predicted

Table 2: Ladder B $c_{n,0}^{(B)}/\nu$ exact fractions ($n = 5$–14).

n	PN	$c_{n,0}^{(B)}/\nu$ (exact)	Numerical	Status
5	4PN	$-63707/1440$	-44.241	**Confirmed [GR seed]**
6	5PN	$-331340107/944640$	-350.758	Predicted (r_B)
7	6PN	$-1723299896507/619683840$	-2780.934	Predicted
8	7PN	$-8962882761732907/406512599040$	-22048.2	Predicted
9	8PN		-1.748×10^5	Predicted
10	9PN		-1.387×10^6	Predicted
11	10PN		-1.100×10^7	Predicted
12	11PN		-8.712×10^7	Predicted
13	12PN		-6.907×10^8	Predicted
14	13PN		-5.476×10^9	Predicted

4 The General $c_{n,k}$ Formula

Theorem 4.1 (General $c_{n,k}$). *The complete transcendental coefficient of $\nu^{k+1}u^n$ in $A(u;\nu)$ is, for all $n \geq 4$ and $k = 0, \ldots, n-4$:*

$$\boxed{c_{n,k}^{\text{trans}} = (-4)^k \left[\left(-\frac{41}{32}\right) \left(\frac{5201}{656}\right)^{n-4} \pi^{2(n-3)} + \left(-\frac{63707}{1440}\right) \left(\frac{5201}{656}\right)^{n-5} \pi^{2(n-4)}\, \Theta(n-5) \right]} \tag{3}$$

where $\Theta(n-5) = 1$ for $n \geq 5$, 0 for $n = 4$.

This factors as:

$$c_{n,k}^{\text{trans}} = (-4)^k \cdot \left(\frac{5201}{656}\right)^{n-5} \pi^{2(n-4)} \cdot \Lambda, \quad \Lambda \equiv -\frac{41 \cdot 5201}{32 \cdot 656}\pi^2 - \frac{63707}{1440} = -7.831 \quad (n \geq 5). \tag{4}$$

Corollary 4.2 (Nu-tower ratio)**.**

$$\frac{c_{n,k}}{c_{n,k-1}} = -4 \quad \textit{for all } k \geq 1, \quad n \geq 5. \tag{5}$$

Table 3: $c^A_{n,k}$ (Ladder A component only, exact fractions, $n = 4$–9, $k = 0$–5). Each row k satisfies $c_{n,k} = (-4)^k c_{n,0}$.

n	PN	$k=0$	$k=1$	$k=2$	$k=3$	$k=4$
4	3PN	−1.2813	—	—	—	—
5	4PN	−10.158	+40.633	—	—	—
6	5PN	−80.538	+322.151	−1288.605	—	—
7	6PN	−638.532	+2554.129	−10216.518	+40866.072	—
8	7PN	−5062.5	+20250.0	−81000.2	+324000.7	−1296002.7
9	8PN	−40131.9	+160527.7	−642110.7	+2568442.9	−10273771.7

5 Combined Transcendentals and the Dual-Ladder Table

Table 4: Combined transcendental contributions to a_n^{trans}/ν ($k = 0$). The ratio $A/B \to 5201\pi^2/656 \approx 78.3$ as $n \to \infty$.

n	PN	Ladder A/nu	Ladder B/nu	Total trans/nu	Status
4	3PN	−12.645	0	−12.645	**Exact**
5	4PN	−989.501	−436.641	−1426.142	**Exact**
6	5PN	−77428.2	−34167.0	−111595.2	Predicted
7	6PN	-6.059×10^6	-2.674×10^6	-8.732×10^6	Predicted
8	7PN	-4.741×10^8	-2.092×10^8	-6.833×10^8	Predicted
9	8PN	-3.710×10^{10}	-1.637×10^{10}	-5.347×10^{10}	Predicted
10	9PN	-2.903×10^{12}	-1.281×10^{12}	-4.184×10^{12}	Predicted
11	10PN	-2.272×10^{14}	-1.002×10^{14}	-3.274×10^{14}	Predicted
12	11PN	-1.777×10^{16}	-7.843×10^{15}	-2.562×10^{16}	Predicted
13	12PN	-1.391×10^{18}	-6.137×10^{17}	-2.005×10^{18}	Predicted
14	13PN	-1.088×10^{20}	-4.803×10^{19}	-1.569×10^{20}	Predicted
15	14PN	-8.516×10^{21}	-3.758×10^{21}	-1.227×10^{22}	Predicted
16	15PN	-6.664×10^{23}	-2.941×10^{23}	-9.605×10^{23}	Predicted
17	16PN	-5.214×10^{25}	-2.301×10^{25}	-7.516×10^{25}	Predicted
18	17PN	-4.080×10^{27}	-1.801×10^{27}	-5.881×10^{27}	Predicted
19	18PN	-3.193×10^{29}	-1.409×10^{29}	-4.602×10^{29}	Predicted
20	19PN	-2.498×10^{31}	-1.102×10^{31}	-3.601×10^{31}	Predicted

Remark 5.1. The ratio Ladder A / Ladder B converges to:

$$\frac{c_{n,0}^{(A)}}{c_{n,0}^{(B)}} \xrightarrow{n\to\infty} \frac{(-41/32)}{(-63707/1440)} \cdot \frac{5201}{656} \cdot \pi^2 = \frac{41 \times 1440 \times 5201}{32 \times 63707 \times 656}\,\pi^2 \approx 2.269 \cdot \pi^2 \approx 22.39. \tag{6}$$

Both ladders have the same asymptotic growth; Ladder A is ≈ 22.4 times larger.

6 The Closed-Form Transcendental Sector

Theorem 6.1 (Resummed transcendental sector)**.** *The complete transcendental sector of $A(u;\nu)$ (both ladders, all ν-orders) is the rational function:*

$$\boxed{A_{\text{trans}}(u;\nu) = \frac{\nu\, u^4\Big[-\dfrac{41\pi^2}{32} - \dfrac{63707\pi^2}{1440}\,u\Big]}{\Big(1 - \dfrac{5201\pi^2}{656}\,u\Big)(1-(1-4\nu)\nu)}} \tag{7}$$

valid for $|u| < 656/(5201\pi^2) \approx 0.01278$ and $\nu \le \frac{1}{4}$.

Equivalently:

$$A_{\text{trans}}(u;\nu) = \frac{-\pi^2\nu u^4\,\Lambda'}{(1-\xi u)(1-\rho)}, \quad \Lambda' = \frac{41}{32} + \frac{63707}{1440}\,u, \quad \xi = \frac{5201\pi^2}{656}, \quad \rho = (1-4\nu)\nu. \tag{8}$$

Remark 6.2 (Convergence). The pole at $u = 656/(5201\pi^2) \approx 0.01278$ lies well inside the light-ring $u = 1/3$. This is physically expected: the transcendental sector alone does not converge at the ISCO. The EOB resummation (Padé approximant of the full A) extends the radius to $u < 1/2$.

7 Complete $A(u;\nu)$: All Sectors

$$\boxed{A(u;\nu) = \underbrace{1-2u}_{\text{Schwarzschild}} + \underbrace{2\nu u^3}_{\text{2PN}} + \underbrace{A_{\text{trans}}(u;\nu)}_{\text{Eq. (7), exact}} + \underbrace{R(u;\nu)}_{\text{rational, partial}} + \underbrace{L(u;\nu)}_{\text{log/tail, partial}}} \tag{9}$$

Known exact rational terms:

$$R(u;\nu) \supset \nu\left(\frac{94}{3}u^4 + \frac{331054}{175}u^5\right) + \nu^2\left(-\frac{221}{6}u^5\right) + \cdots \tag{10}$$

Known exact log terms:

$$L(u;\nu) \supset \frac{64}{5}\nu u^5 \ln u + \left(\frac{1208}{105} - \frac{64}{5}\ln 4\right)\nu u^6 \ln u + \cdots \tag{11}$$

8 Verification Against GR

8.1 4PN exact decomposition (zero SymPy residual)

$$c_{5,1} = \underbrace{\frac{331054}{175}}_{\approx +1891.7} + \underbrace{\frac{c_{5,0}^{(B)}}{\nu} \cdot \pi^2}_{-436.641} + \underbrace{\frac{c_{5,0}^{(A)}}{\nu} \cdot \pi^4}_{-989.501} = +465.595 \tag{12}$$

SymPy residual: identically zero.

8.2 Full comparison table

Table 5: QGD vs. GR (March 2026). All items trace to a Theorem 1.1 rule.

Coefficient	GR value	QGD origin	Status
$c_{3,1}$	2.000	Rule R2	**Exact**
$c_{4,1}$	18.688	Rules R2+R3	**Exact**
$c_{4,1}$ (π^2 coeff)	$-41/32$	Rule R3, 1-loop	**Exact**
$c_{5,1}$ (π^4 coeff)	$-5201/512$	Rule R3, 2-loop	**Exact**
$c_{5,1}$ (π^2 coeff)	$-63707/1440$	Rule R4, G_{m_Q} seed	**Confirmed**
$c_{5,1}$ (total, Fokker)	$+465.595$	All three parts	**Exact**
$c_{5,2}$ (π^2 coeff)	$+41/32$	Rule R5, $k = 1$	**Exact**
$a_{5\log}/\nu$	$64/5$	Rule R6, G_{m_Q} tail	**Exact**
$a_{6\log}/\nu$	-6.240	Rule R6, tail-of-tail	**Confirmed**
$c_{6,A}/\nu$ (π^6)	—	$-27050401/335872 \approx -80.5$	**Predicted**
$c_{6,B}/\nu$ (π^4)	—	$-331340107/944640 \approx -350.8$	Predicted (r_B)
$c_{n,k}^{\text{trans}}$, $n \geq 6$	—	Eq. (3)	Predicted
$c_{5,1}^{\text{rat}}$, $c_{5,2}^{\text{rat}}$	$+1891.7$, -36.8	Fokker integrals	Known from GR
$c_{n,k}^{\text{rat}}$, $n \geq 6$	—	5-loop Fokker	Open
$c_{7,3}$ (4 unknowns)	—	3-body integral	Open (GR frontier)

9 EOB Binding Energy: Exact Predictions

From $a_3 = 2\nu$ and $a_4 = \nu(94/3 - 41\pi^2/32)$ alone (SymPy zero residual):

$$E_{\text{bind}}[u^6]_{\nu^3} = -\frac{6699}{1024} + \frac{123\pi^2}{512} \approx -4.1710, \tag{13}$$

$$E_{\text{bind}}[u^6]_{\nu^4} = -\frac{55}{1024} \approx -0.0537, \tag{14}$$

$$E_{\text{bind}}[u^6]_{\nu^5} = -\frac{21}{1024} \approx -0.0205. \tag{15}$$

10 Convergence Analysis

The ratio test on A_{trans}:

$$\left|\frac{a_{n+1}^{\text{trans}}}{a_n^{\text{trans}}}\right| u \xrightarrow{n\to\infty} \frac{5201\pi^2}{656}\, u \approx 78.3\, u. \tag{16}$$

Convergence requires $u < 656/(5201\pi^2) \approx 0.01278$.

Physical interpretation. This tight convergence radius is not pathological; it is the standard behaviour of PN series, which are asymptotic (not convergent) expansions. The individual PN orders are large, but they cancel with the rational sector to give the physical a_n values. The EOB Padé resummation restores convergence up to the horizon $u < 1/2$.

The large-cancellation pattern. At each order, the transcendental total is $\sim -1426\, u^5$ at 4PN, while the rational part is $\sim +1892\, u^5$. The $a_5 \sim -45$ is the small difference of two large numbers—a characteristic of gravitational PN series near the light-ring, not an artefact of QGD.

11 Completeness Theorems

Theorem 11.1 (Banach). *$\delta\sigma$ exists uniquely in $C^2(\mathbb{R}^3 \setminus \{r_1, r_2\})$; PN series converges absolutely for $\epsilon = GM/(Rc^2) < 1/C$.*

Theorem 11.2 (Stone). *σ-field evolution is unitary, deterministic, norm-preserving, and time-reversible on $L^2(\mathcal{M}, \sqrt{-g}\, dV)$.*

Theorem 11.3 (Diffeomorphism). *$\Phi : \sigma \mapsto g_{\mu\nu} = \mathcal{M}_{\mu\nu}(\sigma)$ is a C^∞-diffeomorphism* $\text{Sol}_{\text{QGD}} \leftrightarrow \text{Sol}_{\text{GR}}$.

12 Summary: What is Exact, Predicted, and Open

Exact (zero-residual): a_1–a_4, log terms, Ladder A seed and ratio, Ladder B seed, nu-tower formula, 4PN three-part decomposition, EOB binding energy at ν^3, ν^4, ν^5.

Predicted (no free parameters, falsifiable): $c_{n,k}^{\text{trans}}$ for all $n \geq 5$, all k via Eq. (3); Ladder B ratio $r_B = 5201/656$; Ladder A and B to 20PN in Tables 1–4.

Open: Rational sector R_n for $n \geq 6$ (Fokker 5-loop+); Ladder B seed derivation from QGD two-loop G_{m_Q} integral; 4 unknown ν^3 rationals in $c_{7,3}$ (GR frontier); 2275/512 identification in $Q(u;\nu)$ sector.

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A Exact Fractions: Ladder A and B to $n = 11$

Table 6: Exact fraction $c_{n,A}/\nu$ and $c_{n,B}/\nu$ to $n = 11$.

n	$c_{n,A}/\nu$ (exact fraction)	n	$c_{n,B}/\nu$ (exact fraction)
4	$-41/32$	5	$-63707/1440$
5	$-5201/512$	6	$-331340107/944640$
6	$-27050401/335872$	7	$-1723299896507/619683840$
7	$-140689135601/220332032$	8	$-8962882761732907/406512599040$
8	$-731724194260801/144537812992$	9	(large)
9	(large)	10	(large)
10	(large)	11	(large)
11	(large)		

B General Formula: Single Expression for All Transcendentals

For $n \geq 5$, $k = 0, \ldots, n-4$, the single master expression is:

$$c_{n,k}^{\text{trans}} = (-4)^k \cdot \left(\frac{5201}{656}\right)^{n-5} \cdot \pi^{2(n-4)} \cdot \Lambda, \quad \Lambda = -\frac{41 \times 5201\,\pi^2}{32 \times 656} - \frac{63707}{1440} \approx -7.831 - 44.241 = -52.072. \quad (17)$$

At $n = 4$ ($k = 0$ only): $c_{4,0}^{\text{trans}} = -41\pi^2/32$.